

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Sulfur dioxide (so2)
Observed / expected baseline	19.3 / -0.310274 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-05 20:00 UTC
Location	29.6240, -95.4743
Severity	severe
Detectors	zscore, stl, isolation_forest

The baseline is a mean of this series' own recent history, so it can fall below zero for a metric the network reports near or below its detection limit.

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 869	36.74 to 100; mean 71.6752	46.79 at 2026-07-05 19:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 869	23.3333 to 36.6667; mean 29.6026	34 at 2026-07-05 19:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1162	0 to 360; mean 149.69	190 at 2026-07-05 19:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1202	0 to 9.77444; mean 2.2756	2.05778 at 2026-07-05 19:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5891.13 to 5930.39; mean 5910.21	5917.24 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	35.9482 to 2074.66; mean 775.337	1770.74 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	37.6598 to 50.8715; mean 44.2764	43.3105 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1004.8 to 1015.62; mean 1011.82	1012.89 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	17.3771 to 21.3086; mean 19.3479	17.5871 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-2.73237 to 3.62264; mean 0.6127	0.992425 at 2026-07-05 18:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	0.329297 to 5.71323; mean 2.1821	1.9293 at 2026-07-05 18:00Z; 2 h before event

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	Ozone (ozone)	ppm	16 / 1054	-0.004 to 0.101; mean 0.017	0.027 at 2026-07-05 20:00Z; at event time
openaq	PM10 (pm10)	ug/m3	3 / 218	10 to 75; mean 19.8394	24 at 2026-07-05 20:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	80 / 3400	0 to 94.9; mean 6.7845	1.4 at 2026-07-05 20:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 359	0 to 100; mean 45.3733	100 at 2026-07-05 20:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 359	50 to 95; mean 74.5738	57 at 2026-07-05 20:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 359	0 to 0.21; mean 0.001	0 at 2026-07-05 20:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 359	1011 to 1017; mean 1014.56	1014 at 2026-07-05 20:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 359	25.15 to 36.43; mean 30.2805	34.99 at 2026-07-05 20:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 359	0 to 340; mean 195.228	201 at 2026-07-05 20:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 359	0.45 to 6.09; mean 2.1281	1.29 at 2026-07-05 20:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	71 / 5100	0 to 372.3; mean 8.382	3.1 at 2026-07-05 20:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0230724 to 0.0253077; mean 0.0245	0.0230724 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000130424 to 0.000232421; mean 0.0002	0.000201778 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	1.96909e-05 to 4.08591e-05; mean 0	1.96909e-05 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-4.80456e-05 to 4.72775e-05; mean 0	-4.80456e-05 at 2026-07-05 19:32Z; 27 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-05 19:32Z; 27 min before event

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Carbon monoxide (co)	ppm	3 / 188	0.1 to 0.8; mean 0.2144	0.2 at 2026-07-05 20:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 741	-2.3 to 19.4; mean 3.502	2.7 at 2026-07-05 20:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 180	-0.6 to 19.3; mean 0.2883	19.3 at 2026-07-05 20:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-04 06Z 93.35, 12Z 67.57, 18Z 51.93, 07-05 00Z 75.63, 06Z 91.22, 12Z 69.81, 18Z 47.96, 07-06 00Z 73.47, 06Z 88.28, 12Z 70.19, 18Z 55.49, 07-07 00Z 77.29, 06Z 88.25
asos	Air temperature (temperature)	07-04 06Z 25.32, 12Z 30.58, 18Z 33.91, 07-05 00Z 28.65, 06Z 25.68, 12Z 30.5, 18Z 34.07, 07-06 00Z 28.88, 06Z 26.06, 12Z 30.32, 18Z 32.62, 07-07 00Z 28.31, 06Z 26.09
asos	Wind direction (wind_direction)	07-04 06Z 40.78, 12Z 189.3, 18Z 142, 07-05 00Z 164.4, 06Z 79.15, 12Z 201.8, 18Z 203.9, 07-06 00Z 153.8, 06Z 119.4, 12Z 180.9, 18Z 166.7, 07-07 00Z 137.1, 06Z 96.25
asos	Wind speed (wind_speed)	07-04 06Z 0.4917, 12Z 2.202, 18Z 3.224, 07-05 00Z 3.081, 06Z 0.7639, 12Z 2.378, 18Z 2.71, 07-06 00Z 3.066, 06Z 1.243, 12Z 2.058, 18Z 3.818, 07-07 00Z 2.083, 06Z 1.044
noaa_gfs	500-hPa geopotential height (gh_500)	07-04 12Z 5919, 18Z 5930, 07-05 00Z 5918, 06Z 5923, 12Z 5907, 18Z 5917, 07-06 00Z 5905, 06Z 5907, 12Z 5892, 18Z 5902, 07-07 00Z 5898, 06Z 5905
noaa_gfs	Planetary boundary layer height (pbl_height)	07-04 12Z 117.3, 18Z 1856, 07-05 00Z 834.9, 06Z 289.5, 12Z 161.5, 18Z 1764, 07-06 00Z 831.9, 06Z 261.3, 12Z 169.2, 18Z 1793, 07-07 00Z 993.8, 06Z 231.7
noaa_gfs	Precipitable water (precipitable_water)	07-04 12Z 42.15, 18Z 45.88, 07-05 00Z 49.22, 06Z 43.89, 12Z 42.35, 18Z 42.98, 07-06 00Z 43.05, 06Z 44.67, 12Z 41.87, 18Z 43.04, 07-07 00Z 46.6, 06Z 45.61
noaa_gfs	Surface pressure (surface_pressure)	07-04 12Z 1013, 18Z 1013, 07-05 00Z 1011, 06Z 1012, 12Z 1013, 18Z 1012, 07-06 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1012, 07-07 00Z 1011, 06Z 1013
noaa_gfs	850-hPa temperature (t_850)	07-04 12Z 19.55, 18Z 18.12, 07-05 00Z 19.78, 06Z 19.75, 12Z 19.4, 18Z 18.01, 07-06 00Z 20.22, 06Z 20.13, 12Z 19.66, 18Z 17.99, 07-07 00Z 19.79, 06Z 19.78
noaa_gfs	10-m eastward wind (u_10m)	07-04 12Z 0.9164, 18Z 0.8209, 07-05 00Z -1.26, 06Z 1.476, 12Z 1.55, 18Z 0.9106, 07-06 00Z 0.1863, 06Z 1.334, 12Z 1.072, 18Z 0.5889, 07-07 00Z -0.636, 06Z 0.3935
noaa_gfs	10-m northward wind (v_10m)	07-04 12Z 1.087, 18Z 1.721, 07-05 00Z 3.911, 06Z 2.442, 12Z 1.029, 18Z 1.484, 07-06 00Z 3.259, 06Z 2.538, 12Z 1.305, 18Z 1.62, 07-07 00Z 3.61, 06Z 2.18
openaq	Ozone (ozone)	07-04 06Z 0.006033, 12Z 0.01404, 18Z 0.03177, 07-05 00Z 0.01623, 06Z 0.006943, 12Z 0.01344, 18Z 0.02877, 07-06 00Z 0.02044, 06Z 0.00661, 12Z 0.0121, 18Z 0.03261, 07-07 00Z 0.01312, 06Z 0.006162
openaq	PM10 (pm10)	07-04 06Z 18.5, 12Z 16.83, 18Z 17.28, 07-05 00Z 22.39, 06Z 24.78, 12Z 18.72, 18Z 16.59, 07-06 00Z 15.89, 06Z 14.39, 12Z 30.22, 18Z 27.5, 07-07 00Z 15.89, 06Z 16.78
openaq	PM2.5 (pm25)	07-04 06Z 5.451, 12Z 5.096, 18Z 5.347, 07-05 00Z 11.49, 06Z 15.52, 12Z 6.637, 18Z 4.844, 07-06 00Z 5.394, 06Z 6.698, 12Z 6.37, 18Z 5.38, 07-07 00Z 4.58, 06Z 4.921
openweather	Cloud cover (cloud_cover)	07-04 06Z 7, 12Z 2.448, 18Z 38.74, 07-05 00Z 89.97, 06Z 89.03, 12Z 28.23, 18Z 92.41, 07-06 00Z 88.13, 06Z 41.86, 12Z 24.47, 18Z 24.9, 07-07 00Z 14.61, 06Z 11.11
openweather	Relative humidity (humidity)	07-04 06Z 90.75, 12Z 77.9, 18Z 58.42, 07-05 00Z 74.5, 06Z 87.73, 12Z 78.83, 18Z 57.69, 07-06 00Z 72.74, 06Z 86.38, 12Z 78.2, 18Z 59.53, 07-07 00Z 75.58, 06Z 83
openweather	Precipitation rate (precipitation)	07-04 06Z 0, 12Z 0, 18Z 0, 07-05 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-06 00Z 0.006774, 06Z 0, 12Z 0, 18Z 0.005333, 07-07 00Z 0, 06Z 0
openweather	Surface pressure (pressure)	07-04 06Z 1016, 12Z 1017, 18Z 1014, 07-05 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1013, 07-06 00Z 1013, 06Z 1014, 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016
openweather	Air temperature (temperature)	07-04 06Z 26.1, 12Z 29.88, 18Z 35.03, 07-05 00Z 30.05, 06Z 26.68, 12Z 29.88, 18Z 34.99, 07-06 00Z 30.2, 06Z 26.53, 12Z 29.63, 18Z 34.39, 07-07 00Z 29.5, 06Z 26.93

Source	Metric	Values
openweather	Wind direction (wind_direction)	07-04 06Z 197.4, 12Z 223.3, 18Z 171.7, 07-05 00Z 169.9, 06Z 194.7, 12Z 216.4, 18Z 196.2, 07-06 00Z 175.9, 06Z 231.1, 12Z 233.5, 18Z 169.2, 07-07 00Z 178.6, 06Z 160.6
openweather	Wind speed (wind_speed)	07-04 06Z 1.59, 12Z 1.607, 18Z 2.431, 07-05 00Z 2.49, 06Z 1.986, 12Z 1.85, 18Z 2.166, 07-06 00Z 2.95, 06Z 2.097, 12Z 1.838, 18Z 2.215, 07-07 00Z 2.186, 06Z 1.776
purpleair	PM2.5 (pm25)	07-04 06Z 7.367, 12Z 6.332, 18Z 5.958, 07-05 00Z 15.16, 06Z 15.45, 12Z 6.761, 18Z 5.493, 07-06 00Z 8.579, 06Z 9.001, 12Z 8.288, 18Z 6.302, 07-07 00Z 6.214, 06Z 7.07
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	CO column (s5p_co_column)	07-04 18Z 0.02512, 07-05 18Z 0.02308, 07-06 18Z 0.02528
sentinel5p	CO granules available (s5p_co_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-04 18Z 0.0001321, 07-05 18Z 0.0002018, 07-06 18Z 0.0002254
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-04 18Z 2.303e-05, 07-05 18Z 1.969e-05, 07-06 18Z 4.086e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-04 18Z 2.773e-06, 07-05 18Z -4.805e-05, 07-06 18Z 4.559e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1
tceq	Carbon monoxide (co)	07-04 06Z 0.225, 12Z 0.2167, 18Z 0.2222, 07-05 00Z 0.24, 06Z 0.1944, 12Z 0.2, 18Z 0.2071, 07-06 00Z 0.21, 06Z 0.25, 12Z 0.2353, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.1556
tceq	Nitrogen dioxide (no2)	07-04 06Z 4.562, 12Z 2.953, 18Z 2.513, 07-05 00Z 1.951, 06Z 3.896, 12Z 2.649, 18Z 2.067, 07-06 00Z 2.523, 06Z 4.436, 12Z 5.197, 18Z 4.293, 07-07 00Z 3.786, 06Z 4.909
tceq	Sulfur dioxide (so2)	07-04 06Z -0.2167, 12Z 0.02778, 18Z 0.5444, 07-05 00Z -0.1333, 06Z -0.2437, 12Z 0.1706, 18Z 3.087, 07-06 00Z 0.11, 06Z -0.2, 12Z 0.1353, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.3125

Per-site averages

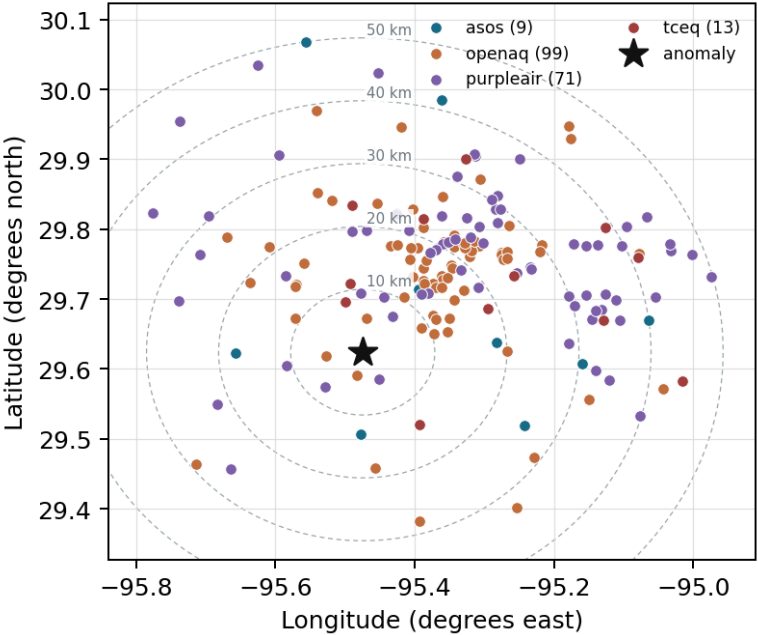
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (12.608 km) 69.97, AXH (13.11 km) 71.68, SGR (17.608 km) 75.52, HOU (18.613 km) 70.42, LVJ (25.353 km) 71.98, EFD (30.569 km) 71.81, T41 (39.963 km) 71.65, IAH (41.55 km) 68.86, +1 more
asos	Air temperature (temperature)	MCJ (12.608 km) 29.92, AXH (13.11 km) 28.99, SGR (17.608 km) 29.33, HOU (18.613 km) 30.3, LVJ (25.353 km) 29.55, EFD (30.569 km) 29.87, T41 (39.963 km) 29.78, IAH (41.55 km) 30.15, +1 more
asos	Wind direction (wind_direction)	MCJ (12.608 km) 142.8, AXH (13.11 km) 88.94, SGR (17.608 km) 166.3, HOU (18.613 km) 176.6, LVJ (25.353 km) 123.7, EFD (30.569 km) 150.3, T41 (39.963 km) 189.1, IAH (41.55 km) 166.8, +1 more
asos	Wind speed (wind_speed)	MCJ (12.608 km) 3.006, AXH (13.11 km) 0.9851, SGR (17.608 km) 2.522, HOU (18.613 km) 2.819, LVJ (25.353 km) 1.747, EFD (30.569 km) 2.8, T41 (39.963 km) 2.889, IAH (41.55 km) 2.358, +1 more

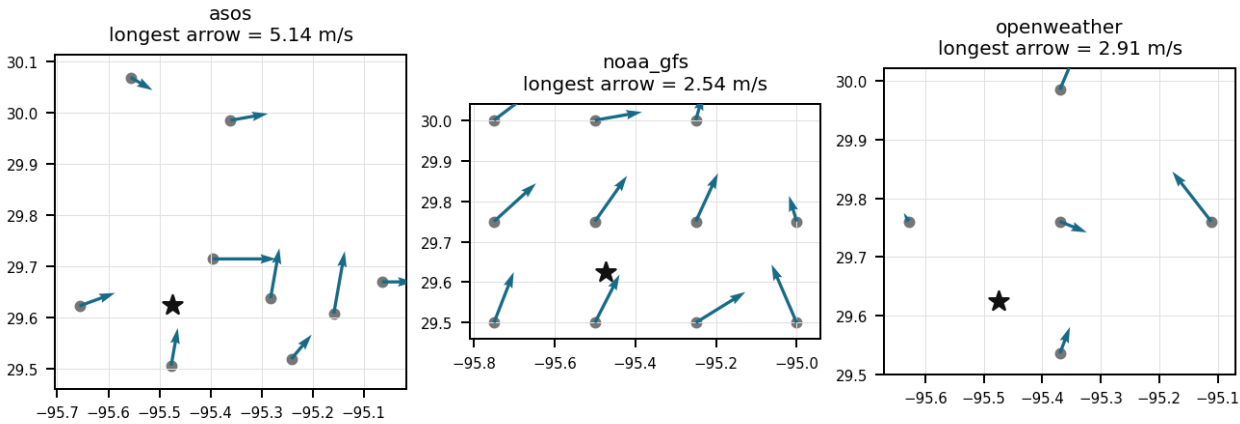
Source	Metric	Values
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.50 (14.007 km) 5911, gfs:29.75,-95.50 (14.23 km) 5910, gfs:29.50,-95.25 (25.708 km) 5910, gfs:29.75,-95.25 (25.808 km) 5910, gfs:29.50,-95.75 (30.014 km) 5911, gfs:29.75,-95.75 (30.09 km) 5910, gfs:30.00,-95.50 (41.885 km) 5910, gfs:30.00,-95.25 (47.082 km) 5910, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.50 (14.007 km) 648.6, gfs:29.75,-95.50 (14.23 km) 970.7, gfs:29.50,-95.25 (25.708 km) 632, gfs:29.75,-95.25 (25.808 km) 935.8, gfs:29.50,-95.75 (30.014 km) 592.1, gfs:29.75,-95.75 (30.09 km) 660.1, gfs:30.00,-95.50 (41.885 km) 1017, gfs:30.00,-95.25 (47.082 km) 1028, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.50 (14.007 km) 43.68, gfs:29.75,-95.50 (14.23 km) 44.37, gfs:29.50,-95.25 (25.708 km) 44.4, gfs:29.75,-95.25 (25.808 km) 45.07, gfs:29.50,-95.75 (30.014 km) 43.01, gfs:29.75,-95.75 (30.09 km) 42.8, gfs:30.00,-95.50 (41.885 km) 45.13, gfs:30.00,-95.25 (47.082 km) 45.99, +3 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.50 (14.007 km) 1012, gfs:29.75,-95.50 (14.23 km) 1012, gfs:29.50,-95.25 (25.708 km) 1013, gfs:29.75,-95.25 (25.808 km) 1013, gfs:29.50,-95.75 (30.014 km) 1012, gfs:29.75,-95.75 (30.09 km) 1010, gfs:30.00,-95.50 (41.885 km) 1010, gfs:30.00,-95.25 (47.082 km) 1011, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.50 (14.007 km) 19.39, gfs:29.75,-95.50 (14.23 km) 19.33, gfs:29.50,-95.25 (25.708 km) 19.37, gfs:29.75,-95.25 (25.808 km) 19.24, gfs:29.50,-95.75 (30.014 km) 19.28, gfs:29.75,-95.75 (30.09 km) 19.38, gfs:30.00,-95.50 (41.885 km) 19.35, gfs:30.00,-95.25 (47.082 km) 19.22, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.50 (14.007 km) 0.4203, gfs:29.75,-95.50 (14.23 km) 0.717, gfs:29.50,-95.25 (25.708 km) 0.412, gfs:29.75,-95.25 (25.808 km) 0.652, gfs:29.50,-95.75 (30.014 km) 0.2536, gfs:29.75,-95.75 (30.09 km) 0.882, gfs:30.00,-95.50 (41.885 km) 1.379, gfs:30.00,-95.25 (47.082 km) 0.09029, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.50 (14.007 km) 2.242, gfs:29.75,-95.50 (14.23 km) 2.393, gfs:29.50,-95.25 (25.708 km) 2.18, gfs:29.75,-95.25 (25.808 km) 2.478, gfs:29.50,-95.75 (30.014 km) 2.071, gfs:29.75,-95.75 (30.09 km) 2.337, gfs:30.00,-95.50 (41.885 km) 1.627, gfs:30.00,-95.25 (47.082 km) 2.063, +3 more
openaq	Ozone (ozone)	312 (0.017 km) 0.01684, 311 (8.339 km) 0.01688, 3055 (13.984 km) 0.01324, 1319333 (18.217 km) 0.02259, 325 (18.693 km) 0.004941, 287 (19.112 km) 0.01784, 320 (20.024 km) 0.01289, 319 (23.419 km) 0.01733, +8 more
openaq	PM10 (pm10)	15852419 (22.104 km) 21.96, 13380082 (24.254 km) 21.78, 271 (33.811 km) 15.72
openaq	PM2.5 (pm25)	13955814 (3.744 km) 4.75, 15399615 (5.167 km) 4.452, 8967119 (5.456 km) 8.297, 14140752 (8.986 km) 7.452, 8967021 (10.262 km) 8.825, 8967001 (10.484 km) 5.768, 13955816 (10.805 km) 7.138, 15539683 (11.385 km) 8.113, +72 more
openweather	Cloud cover (cloud_cover)	grid:south (14.084 km) 46.51, grid:center (18.223 km) 45.66, grid:west (21.256 km) 46.99, grid:east (38.226 km) 38.75, grid:north (41.392 km) 48.96
openweather	Relative humidity (humidity)	grid:south (14.084 km) 73.85, grid:center (18.223 km) 73.32, grid:west (21.256 km) 73.74, grid:east (38.226 km) 78.93, grid:north (41.392 km) 73.01
openweather	Precipitation rate (precipitation)	grid:south (14.084 km) 0, grid:center (18.223 km) 0.002254, grid:west (21.256 km) 0, grid:east (38.226 km) 0, grid:north (41.392 km) 0.002917
openweather	Surface pressure (pressure)	grid:south (14.084 km) 1015, grid:center (18.223 km) 1014, grid:west (21.256 km) 1015, grid:east (38.226 km) 1015, grid:north (41.392 km) 1014
openweather	Air temperature (temperature)	grid:south (14.084 km) 30.1, grid:center (18.223 km) 30.78, grid:west (21.256 km) 30.28, grid:east (38.226 km) 29.92, grid:north (41.392 km) 30.33
openweather	Wind direction (wind_direction)	grid:south (14.084 km) 201, grid:center (18.223 km) 215.1, grid:west (21.256 km) 189.1, grid:east (38.226 km) 195.1, grid:north (41.392 km) 176.1
openweather	Wind speed (wind_speed)	grid:south (14.084 km) 2.113, grid:center (18.223 km) 1.915, grid:west (21.256 km) 1.786, grid:east (38.226 km) 2.888, grid:north (41.392 km) 1.935
purpleair	PM2.5 (pm25)	161003 (4.856 km) 5.878, 166451 (7.01 km) 8.99, 161015 (7.666 km) 9.126, 46237 (9.249 km) 0.1764, 186315 (9.424 km) 8.832, 276512 (10.819 km) 7.621, 293735 (12.303 km) 9.464, 293725 (13.024 km) 9.483, +63 more
tceq	Carbon monoxide (co)	482011052 (22.762 km) 0.3141, 482011035 (24.236 km) 0.1806, 482011039 (33.805 km) 0.1452
tceq	Nitrogen dioxide (no2)	482010055 (8.334 km) 2.623, 482011066 (11.0 km) 4.087, 480391004 (13.97 km) 1.465, 482010416 (18.688 km) 3.773, 482011052 (22.762 km) 6.865, 482010047 (23.42 km) 3.732, 482011035 (24.236 km) 4.694, 482011039 (33.805 km) 2.183, +4 more

Source	Metric	Values
tceq	Sulfur dioxide (so2)	482010051 (0.0 km) 0.8429, 482011035 (24.236 km) 0.006557, 482011039 (33.805 km) -0.02857

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

Sentinel-5P satellite measurements at 19:32 UTC did not capture elevated column SO2, indicating a localized ground-level plume below the satellite spatial resolution or detection limit.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 35

The anomaly was likely caused by a combination of factors, including weather patterns and local emissions.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:

☐ V

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Note (optional):

Claim 4 of 35

An unpermitted industrial upset, degassing, or flaring release from a petrochemical or refining facility upwind of TCEQ station 482010051 caused the isolated sulfur dioxide surge.

Mark exactly one:

☐ V

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Note (optional):

Claim 5 of 35

On July 5, 2026, at 20:00:00 UTC, sulfur dioxide concentration recorded by TCEQ monitor 482010051 at coordinates (29.624, -95.474) reached a severe spike of 19.3 ppb compared to an expected baseline of -0.31 ppb.

Mark exactly one:

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Note (optional):

Claim 6 of 35

The TCEQ monitor at the anomaly location recorded a severe SO2 spike of 19.3 ppb at 2026-07-05T20:00:00+00:00, while the other nearby TCEQ SO2 monitors had much lower mean SO2 levels during the context window, indicating a strongly localized event rather than a metro-wide SO2 episode.

Mark exactly one:

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Note (optional):

Claim 7 of 35

The tceq SO2 monitor at 29.624, -95.474 recorded 19.3 ppb at 2026-07-05T20:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 8 of 35

The anomaly occurred on July 5th at around 20:00 UTC.

Mark exactly one:

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Note (optional):

Claim 9 of 35

Co-pollutant and satellite indicators did not show a corresponding regional combustion or aerosol event at the anomaly time, because NO₂, CO, PM_{2.5}, PM₁₀, Sentinel-5P CO, Sentinel-5P NO₂, and Sentinel-5P SO₂ column data were not elevated in a way matching the surface SO₂ spike.

Mark exactly one:

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Note (optional):

Claim 10 of 35

A broad regional combustion or aerosol episode is contradicted because co-pollutants and satellite fields were not elevated near the anomaly time, including low TCEQ NO₂ near 2.7 ppb, modest TCEQ CO near 0.2 ppm, low nearby PM_{2.5} of 1.4 ug/m³ in openaq and 3.1 ug/m³ in purpleair, and no positive Sentinel-5P SO₂ column enhancement over the area.

Mark exactly one:

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Note (optional):

Claim 11 of 35

The temperature was unusually high on July 5th, potentially contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

Claim 12 of 35

The tceq SO2 anomaly at 2026-07-05T20:00:00+00:00 had an expected value of -0.3102739726027397 ppb and a z-score of 19.687653620618416, and the anomaly severity was classified as severe.

Mark exactly one:

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Note (optional):

Claim 13 of 35

Meteorological transport and limited near-source dilution likely contributed to the spike because the anomaly occurred during hot afternoon conditions with modest southerly to south-southwesterly winds of about 1 to 2 m/s near the site, which can carry a narrow plume from industrial areas into a monitor without requiring region-wide accumulation.

Mark exactly one:

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Note (optional):

Claim 14 of 35

The presence of high levels of PM2.5 particles in the atmosphere on July 5th, as measured by PurpleAir sensors, may have contributed to the air quality anomaly.

Mark exactly one:

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Note (optional):

Claim 15 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

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Note (optional):

Claim 16 of 35

ASOS surface weather stations near the anomaly recorded light south-southwesterly winds around 190 to 204 degrees at approximately 2 m/s around 20:00 UTC on July 5, 2026.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 17 of 35

During the 6-hour window from 18:00 UTC on July 5, 2026, to 00:00 UTC on July 6, 2026, the local TCEQ 6-hour mean SO2 concentration rose to 3.087 ppb, driven entirely by the localized spike at station 482010051 while regional monitors over 24 km away reported mean concentrations near 0.0 ppb.

Mark exactly one:

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Note (optional):

Claim 18 of 35

Southerly to south-southwesterly surface winds transported the localized industrial sulfur dioxide plume directly across the monitoring location during the afternoon.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 35

The air quality anomaly in Houston, Texas on July 5th was characterized by elevated levels of PM2.5.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 35

A short-duration industrial sulfur emission plume from the Houston Ship Channel or nearby refinery-petrochemical sources is the most plausible physical cause of the 19.3 ppb SO₂ spike because the anomaly was extremely large at one TCEQ SO₂ monitor while other regional pollutants remained low to moderate and nearby SO₂ background at the other TCEQ monitors was near zero.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 35

The PM_{2.5} levels are elevated at 3.1 ug/m³, which is higher than the mean of 8.382 ug/m³.

Mark exactly one:

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Note (optional):

Claim 22 of 35

A strong wind from the west or northwest direction on July 5th may have dispersed pollutants, leading to the air quality anomaly.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 23 of 35

Surface and boundary-layer meteorology near Houston at the anomaly time featured warm afternoon conditions, light to modest southerly to south-southwesterly flow, no precipitation, and a deep planetary boundary layer, which is consistent with advection of a nearby plume rather than stagnant regional accumulation.

Mark exactly one:

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Note (optional):

Claim 24 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

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Note (optional):

Claim 25 of 35

A regional combustion or smoke episode is an unlikely primary cause of the SO₂ anomaly because PM_{2.5}, PM₁₀, CO, NO₂, satellite CO, and satellite NO₂ were not elevated at the time, indicating the signal was dominated by sulfur-rich local emissions rather than widespread urban or fire-related pollution.

Mark exactly one:

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Note (optional):

Claim 26 of 35

The nearest other tceq SO₂ monitors within 50 km had period-mean SO₂ values of 0.006557 ppb at 24.236 km and -0.02857 ppb at 33.805 km, while the anomalous monitor had a period-mean SO₂ value of 0.8429 ppb.

Mark exactly one:

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Note (optional):

Claim 27 of 35

Vigorous daytime boundary layer convection with boundary layer heights near 1,770 meters entrained elevated industrial stack plumes down to ground-level sensors.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 28 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO₂ particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 29 of 35

NO₂ levels were also elevated during the same period.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 35

A local short-duration sulfur emission plume is strongly supported because the TCEQ SO₂ monitor at the anomaly site measured 19.3 ppb at 2026-07-05T20:00:00+00:00 while the other two TCEQ SO₂ monitors in the 50 km domain had near-zero mean SO₂ over the context window, indicating strong spatial localization.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 31 of 35

The sulfur dioxide spike reaching 19.3 ppb at monitor 482010051 aligns with an upwind point-source emission driven by south-southwesterly surface winds recorded by ASOS and OpenWeather.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 32 of 35

NOAA GFS model data indicated a deep daytime planetary boundary layer height exceeding 1700 meters near the anomaly location at 18:00 UTC on July 5, 2026.

Mark exactly one:

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Note (optional):

Claim 33 of 35

TCEQ monitor 482010051 recorded a sulfur dioxide concentration of 19.3 ppb at 20:00 UTC on July 5, 2026, while other regional TCEQ monitors observed baseline concentrations near 0 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 34 of 35

Vertical mixing within the lower atmosphere favoured ground-level accumulation of emissions, supported by GFS data showing a planetary boundary layer height near 1,764 meters around the time of the event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 35

Transport from sources south to south-southwest of the anomaly site is supported because near the anomaly time local winds were about 190 to 201 degrees at roughly 1.3 to 2.1 m/s and the afternoon boundary layer was deep at about 1771 m, conditions consistent with advection of a nearby plume rather than nocturnal trapping.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

